Part 3. Storing data in the cloud

There is one guy named Singleton in your office who knows all about your file server. If Singleton is out of office, no one else can maintain the file server. As you can imagine, while Singleton is on vacation, the file server crashes. No one else knows where the backup is located, but the boss needs a document now or the company will lose a lot of money. If Singleton had stored his knowledge in a database, coworkers could look up the information. But because the knowledge and Singleton are tidily coupled, the information is unavailable.

Imagine a virtual machine where important files are located on hard disk. As long as the virtual machine is up and running, everything is fine. But everything fails all the time, including virtual machines. If a user uploads a document on your website, where is it stored? Chances are high that the document is persisted to hard disk on the virtual machine. Let's imagine that the document was uploaded to your website but persisted as an object in an independent object store. If the virtual machine fails, the document will still be available. If you need two virtual machines to handle the load on your website, they both have access to that document because it is not tightly coupled with a single virtual machine. If you separate your state from your virtual machine, you will be able to become fault tolerant and elastic. Let highly specialized solutions like object stores and databases persist your state.

AWS offers many ways to store your data. The following table can help you decide which service to use for your data at a high level.